Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “kinetic energy” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

kinetic energy: energy associated with motion

scalar: a quantity described by size without a vector direction

01 Warm-up: bring back an earlier idea

A and B have equal momentum magnitude, but A has **3** times B’s mass. **Compare speeds** and kinetic energies algebraically.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar. Neither alone gives a complete damage forecast.

Worked reasoning: A **2 kg** cart at **3 m/s** has momentum **6 kg·m/s** and kinetic energy $\frac{1}{2}\times 2\times 3^2 = 9$ **J**. A **1 kg** cart at **6 m/s** has the same momentum but **18 J** of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal-mass carts move at speeds **2** and **4 m/s**. **Predict how their momenta** and kinetic energies compare before calculating.

Sentence starter: Both … . They differ in … because … .



Same mass; 2 m/s Same mass; 4 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

Two lab carts each have momentum magnitude **12 kg·m/s**. Their masses are **2 kg** and **6 kg**.
1. **Find each speed** and kinetic energy.
2. **Draw energy bars** and **explain why equal momentum** does not guarantee equal kinetic energy.

Word help: magnitude = size of a quantity without its direction

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

1. **List three forms or destinations** of energy after a deforming collision.
2. Does “less kinetic energy” mean “energy destroyed”?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

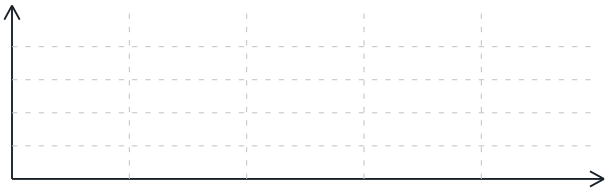
06 Review: connect an earlier idea

A positive triangular force pulse lasts **0.4 s** and peaks at **60 N**. **Find impulse** and velocity change for a **0.8 kg** object.
Show the signed impulse on the force–time graph.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.

Force (N)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

At fixed momentum, mass triples. Derive or explain the kinetic-energy multiplier using $K=p^2/(\mathbf{2\ m})$. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two carts have equal momentum magnitude p. Derive K in terms of p and m, then **compare energies** if one mass is four times the other. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3